Question 1 [CO3] [10 Points]

[Answer on the answer-script]

Write a Java program that takes a string as input from the user. Assume that the input string consists only of lowercase English letters. A letter may appear one or more times in the string. Your task is to create and print a modified string where the first and last occurrences of any letter that appears more than once in the input string are removed. All other occurrences of such letters, as well as letters that appear only once in the input string, should remain in their original order. If the modified string is empty, it should print "No characters left!".

Sample Input	Sample Output
banana	ba
aabbcc	No characters left!
giggling	ggln

Explanation: For the sample input "banana", among the characters 'b', 'a', and 'n', character 'b' occurs only once. That's why it remains unchanged in the modified string.

Characters 'a' and 'n' occur more than once in the input string. The indexes for 'a' are 1, 3, and 5. In the modified string, the first and last occurrences of 'a' (indexes 1 and 5) are removed.

Similarly, for 'n', the indexes are 2 and 4. Therefore, in the modified string, the first and last occurrences of 'n' (indexes 2 and 4) are removed.

Hence, the modified string is "ba".

Question 2 [CO4] [10 Points]

[Answer on the answer-script]

Write a Java program where an array of integers is given. Now, you have to create two new subarrays: one containing all the odd numbers and the other containing all the even numbers. Finally, print both arrays and the difference between the Largest Even Number and the Smallest Odd Number.

Note: You may use Arrays.toString() to print the subarrays.

[Your code should work for any given integer type array. You can assume given array will have at least one odd integer and one even integer.]

Given Array	Sample Output
int [] array = {5, 3, 8, 6, 7, 2, 1,4};	Odd Numbers: [5, 3, 7, 1] Even Numbers: [8, 6, 2, 4] Difference between Largest Even Number and Smallest Odd Number: 7
int [] array = {2, 4, 1, 6, 12};	Odd Numbers: [1] Even Numbers: [2, 4, 6, 12] Difference between Largest Even Number and Smallest Odd Number: 11

Question 3 [CO4] [5 Points]

<u>[Answer on the question paper]</u>

The code below is designed to sort an array in **increasing** order. However, it contains **6 errors**. Analyze the code, identify the errors, and correct them. Provide your corrections in the specified format.

```
public class A{
  public void main(String args []){
    int array = {20, 25, 10, 8, 3};
    for (int i = 0; i <= array.length; i++){
      int min_index = i;
      for (int j = i+1; j < array.length(); j++){
        if (array[j] > array[min_index]){
          min_index = j;
        }
      }
      int temp = array[min_index-1];
      array[min_index] = array[i];
      array[i] = temp;
    }
    System.out.println("Sorted Array in Increasing Order: ");
    for (int j = 0; j < array.length; j++){
      System.out.print(array[j]+" ");
    }
  }
}
```

Write your corrections in the table below. For better understanding, one error correction is shown.

Line Number	Corrected Code Line
Line 2	public static void main(String args []){

Question 4 [CO5] [5 Points]

[Answer on the question paper]

Illustrate the outputs of the following statements. Provide your workings on the answer script to verify your outputs. **Your answer will not be accepted without the workings. All of the outputs must be in the question paper.**

```
public class TracingA {
    public static void main(String[] args) {
        int [] b = {20, 5, 15, 30, 13};
        System.out.println(metA(b, 0));
    }
    public static int metA(int[] a, int i) {
        if (i == a.length - 1) {      (i=4)
            return a[i];
        }
        int x = metA(a, i + 1);
        System.out.println(a[i] + x);
        if (a[i] < x) {
            return a[i];
        } else {
            return x;
        }
    }
}
```

Output